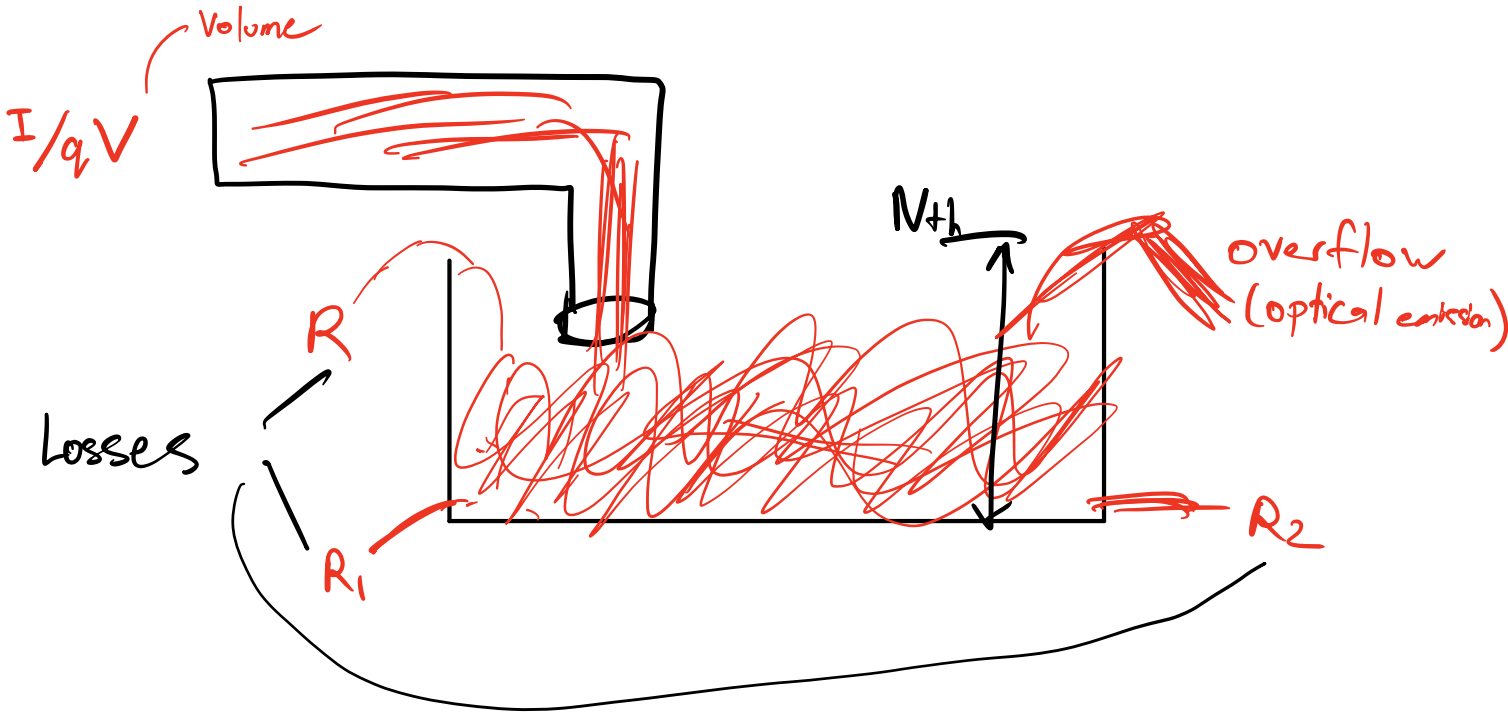
Bathtub Analogy



Laser Rate Equations

- $\frac{dS}{dt}$ – photons

  - LED's → Spontaneous emission
  - Laser's → Stimulated emissions

- $\frac{dN}{dt}$ – electrons

Band Gap

- Direct vs. Indirect Conduction and Valence Bands. Direct is preferred